

Feasibility Analysis: Scaling AI-Based Precision Irrigation from Laboratory Prototype to Field Deployment

Feasibility Analysis Report - CIE-349/408 Embedded Systems

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Abstract—This paper presents a feasibility analysis for scaling a 1-axis laboratory prototype of an AI-based Smart Precision Irrigation System into a 2D agricultural field deployment. The initial prototype successfully integrates edge AI (MobileNetV2 via Raspberry Pi), real-time control (PIC16F877A), and IoT connectivity (ESP8266 to Firebase). However, scaling the mechanical gantry structure to a multi-meter field matrix introduces severe structural and electrical bottlenecks. This report proposes transitioning from a mechanical water-delivery gantry to a distributed IoT sensor network paired with a zone-based solenoid fluidic system. Environmental viability and a complete cost analysis (BOM) for a pilot deployment are also detailed, demonstrating that the revised architecture is highly feasible and economically viable at approximately 9,020 EGP.

Index Terms—Smart Irrigation, Precision Agriculture, Edge AI, IoT, Feasibility Analysis, Embedded Systems.

I. INTRODUCTION

The core cyber-physical loop of the Smart Precision Irrigation System has been successfully validated in a laboratory environment. The system senses local parameters, captures crop images, utilizes a quantized TFLite model to classify plant health (e.g., Healthy, Blight, Pest), and drives a NEMA 17 stepper motor to physically move a water nozzle to the target location along a 1-axis aluminum extrusion rail.

While the computational architecture and the UART/MQTT communication bridges operate flawlessly, extrapolating the 1-axis mechanical design (using 2020 aluminum extrusions and 8 mm smooth rods) to a large 2D field deployment poses significant mechanical, electrical, and economic challenges. This report analyzes these constraints and formalizes the engineering pivots required to deploy the system in a real-world greenhouse or agricultural plot.

II. TECHNICAL FEASIBILITY AND REQUIRED ARCHITECTURAL PIVOTS

A. Mechanical vs. Fluidic Distribution

Constraint: Moving from a 1-axis slider to a 2D matrix requires building a full X-Y gantry system. A 2020 aluminum extrusion and an 8 mm smooth rod cannot span distances greater than 1 to 1.5 meters without experiencing severe deflection (sagging) under their own weight. Adding the payload

of a water-filled hose, a Raspberry Pi 4, and camera equipment will cause linear bearings (LM8UU) to bind. Furthermore, the required holding torque exceeds the capabilities of standard NEMA 17 motors, and timing belts will suffer from stretching and skipped teeth over long spans.

Architectural Pivot: The system must eliminate the physical transportation of the water source.

- *Zone-Based Drip Irrigation:* Stationary PVC pipes and drip lines should be laid across the field, divided into independent irrigation zones.
- *Solenoid Valves:* The NEMA 17 motors and gantry mechanism will be replaced by 12V DC Solenoid Water Valves at the head of each zone. The central PIC16F877A controller will trigger relays to open specific valves based on AI or sensor-driven protocols.

B. Sensing and Power Distribution

Constraint: The current prototype utilizes a single DHT11 sensor and local analog soil moisture sensors. Running 5V or 12V DC lines across a 20+ meter field to power sensors will result in significant voltage drop, rendering analog readings highly inaccurate. Furthermore, the DHT11 is strictly intended for indoor use and will fail under field condensation and dust.

Architectural Pivot: Distributed Wireless Nodes.

- *Soil Nodes:* Deploy individual ESP8266 microcontrollers at each irrigation zone. Each node will locally read its capacitive soil moisture sensor and transmit the data back to the central hub via a wireless mesh or a robust RS-485 wired bus.
- *Weatherproofing:* The environmental sensor must be upgraded to an industrial-grade SHT3x or BME280 sensor, housed within a 3D-printed sintered radiation shield.

C. Edge AI Vision (The Mobile Component)

Since water delivery is now stationary, the Raspberry Pi 4 and Pi Camera must still monitor the crops to execute the MobileNetV2 classifications.

Architectural Pivot: Mount the vision system on an overhead zipline, a drone, or an automated ground rover. This mobile agent operates purely for imaging, pushing detection

coordinates and protocol IDs (0x01 to 0x05) to the Firebase RTDB, allowing the stationary fluidics to actuate accordingly.

III. ENVIRONMENTAL AND MATERIAL FEASIBILITY

Standard 3D-printed PLA parts (used in the laboratory prototype) undergo a glass transition at approximately 60°C. In direct Egyptian sunlight, ambient temperatures inside enclosures will easily exceed this, causing catastrophic warping and structural failure.

All outdoor mechanical mounts and electronic enclosures must be printed using Polyethylene Terephthalate Glycol (PETG) or Acrylonitrile Styrene Acrylate (ASA), both of which offer superior UV resistance and thermal stability. The central control unit housing the PIC and relays must be placed in an off-the-shelf IP65/IP67 rated waterproof enclosure with properly sealed PG7 cable glands.

IV. COST ANALYSIS

The following Bill of Materials (BOM) outlines the estimated costs for scaling the system to a 4-Zone Pilot Deployment in the Egyptian market. The cost of the massive aluminum field gantry is omitted due to the adoption of the fluidic pivot.

TABLE I
ESTIMATED BOM FOR 4-ZONE PILOT DEPLOYMENT

Subsystem	Component	Qty	Total (EGP)
Control Hub	PIC16F877A (Existing)	1	0
	ESP8266 NodeMCU	1	200
	IP65 Waterproof Enclosure	1	350
	12V 30A Power Supply	1	600
	4-Channel 5V Relay	1	120
Fluidics	0.5 HP AC Water Pump	1	2,200
	1/2" 12V Solenoid Valve	4	800
	PVC Pipes & Drip Lines	Lot	800
Sensing	ESP8266 (For nodes)	4	800
	Capacitive Soil Sensor	4	200
	ACS712 Current Sensor	5	400
	SHT30 Temp/Hum Sensor	1	250
Vision (AI)	Raspberry Pi 4 + Camera	1	0
Mechanics	PETG 3D Printing (~200g)	Lot	800
	Rover/Zipline Chassis Kit	1	1,500
Total	Estimated Cost		~9,020

V. CONCLUSION

Scaling the Smart Precision Irrigation System is highly feasible both technically and economically. By decoupling the AI computer vision (rover/camera) from the fluidics (stationary pumps and solenoid valves), the system eliminates the mechanical impossibility of a multi-meter gantry. This updated architecture remains robust against agricultural environments while preserving the sophisticated software, safety logic, and machine learning pipelines developed in the laboratory phase.